

PRACTICAL 14

MINI PROJECT

AIM:

To design and implement a complete database application for a Library Management System by integrating database design with constraints, SQL retrieval and aggregate queries, joins and nested queries, views, transaction control, database security, and PL/SQL programming — including procedures, functions, triggers, and cursors — into a single, cohesive system.

PROJECT OVERVIEW:

The Library Management System is a database application designed to manage book, category, publisher, member, and borrowing information efficiently. The system uses MySQL and MySQL's stored-program features to store, retrieve, and process library records. It integrates relational tables, constraints, SQL queries, joins, aggregate functions, subqueries, views, transaction control, stored procedures, functions, triggers, and cursors. The system helps maintain accurate book inventory, track member activity, manage issue and return records, and generate useful reports.

OBJECTIVE:

- To maintain book, category, publisher, and member information in a centralized database.
- To manage book issue/return records and fines efficiently.
- To retrieve and analyze library data using SQL queries.
- To maintain data consistency using constraints, transactions, and stored procedures.
- To generate useful reports using database features such as joins, views, procedures, functions, triggers, and cursors.

FUTURE SCOPE:

The system can be extended with:

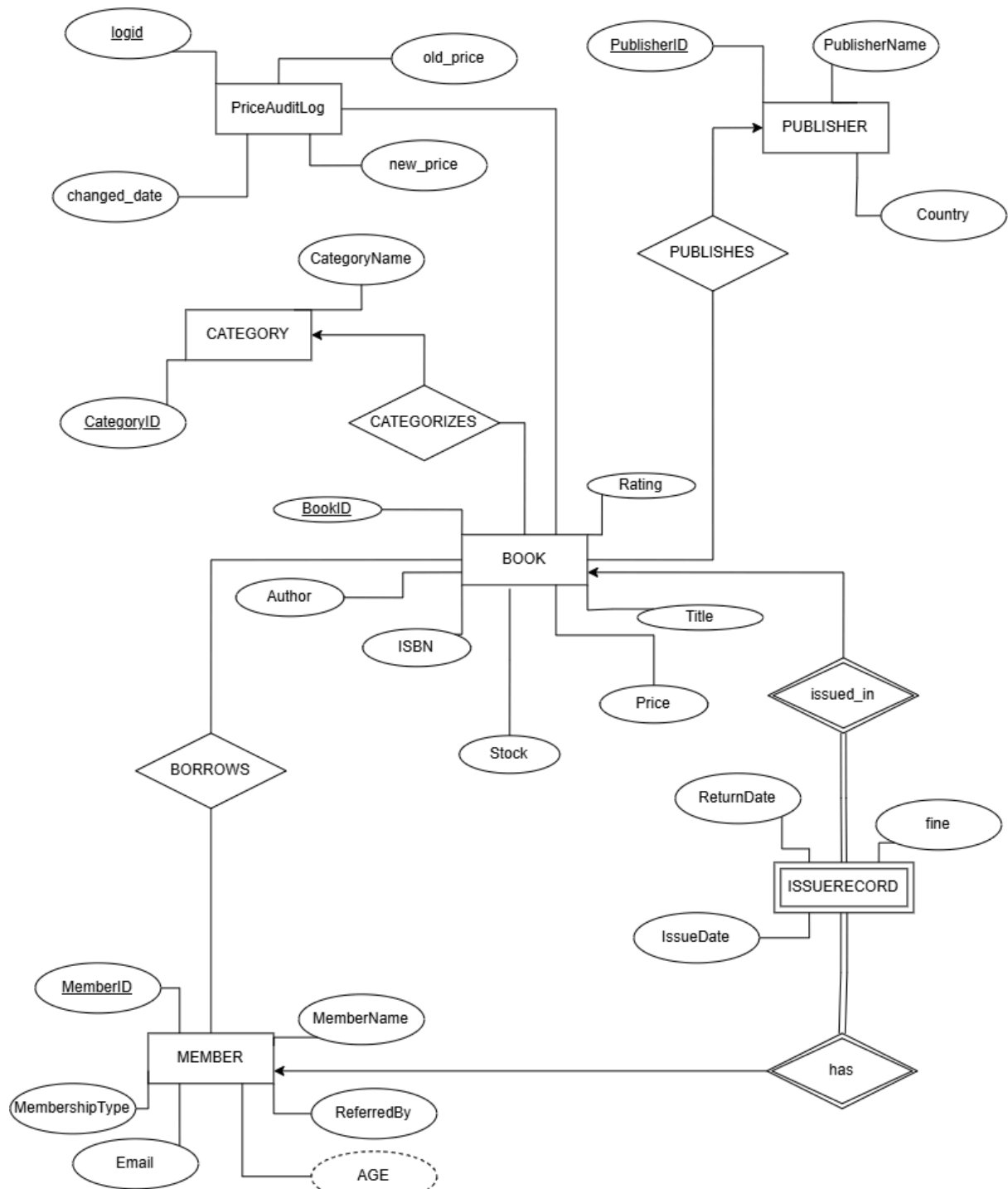
- Online member registration and login
- Fine payment integration
- Book reservation system
- Email/SMS due-date notifications
- Web-based library portal
- Mobile application for members

SOFTWARE USED:*Hardware Requirements*

- Computer/Laptop
- Minimum 4 GB RAM
- Minimum 10 GB free storage

Software Requirements

- MySQL Server
- MySQL Command-Line Client / MySQL Workbench
- SQL and MySQL Stored Programs
- Windows operating system

ER DIAGRAM:

1. DATABASE DESIGN AND CONSTRAINTS:

- **CREATE AND SELECT THE DATABASE**

SYNTAX:

CREATE DATABASE database_name;

USE database_name;

QUERY:

CREATE DATABASE LibraryProjectDB;

USE LibraryProjectDB;

```
mysql> CREATE DATABASE LibraryProjectDB;
Query OK, 1 row affected (0.06 sec)

mysql> USE LibraryProjectDB;
Database changed
mysql>
```

- **CREATE TABLES**

SYNTAX:

CREATE TABLE table_name (

column1 datatype constraint, column2 datatype constraint, ... PRIMARY KEY

(column_name), FOREIGN KEY (column_name) REFERENCES

parent_table(parent_column) ON DELETE {CASCADE | SET NULL} ON UPDATE

{CASCADE | SET NULL});

QUERY:

CREATE TABLE Publisher (PublisherID INT PRIMARY KEY, PublisherName

VARCHAR(60) NOT NULL UNIQUE, Country VARCHAR(40) NOT NULL);

```
CREATE TABLE Category ( CategoryID INT PRIMARY KEY, CategoryName  
VARCHAR(30) NOT NULL UNIQUE );
```

```
CREATE TABLE Book ( BookID INT PRIMARY KEY, Title VARCHAR(100) NOT  
NULL, Author VARCHAR(60), Price DECIMAL(8,2) CHECK (Price > 0), ISBN  
VARCHAR(20) UNIQUE NOT NULL, Rating DECIMAL(2,1), Stock INT CHECK  
(Stock >= 0), CategoryID INT, PublisherID INT, FOREIGN KEY (CategoryID)  
REFERENCES Category(CategoryID) ON DELETE CASCADE ON UPDATE  
CASCADE, FOREIGN KEY (PublisherID) REFERENCES Publisher(PublisherID)  
ON DELETE SET NULL ON UPDATE CASCADE );
```

```
CREATE TABLE Member ( MemberID INT PRIMARY KEY, MemberName  
VARCHAR(60) NOT NULL, Age INT CHECK (Age BETWEEN 12 AND 75),  
Email VARCHAR(100) UNIQUE NOT NULL, MembershipType VARCHAR(10)  
CHECK (MembershipType IN ('Regular','Premium','Student')), ReferredBy INT,  
FOREIGN KEY (ReferredBy) REFERENCES Member(MemberID) );
```

```
CREATE TABLE IssueRecord ( MemberID INT, BookID INT, IssueDate DATE NOT  
NULL, ReturnDate DATE, Fine DECIMAL(6,2) CHECK (Fine >= 0), PRIMARY  
KEY (MemberID, BookID), FOREIGN KEY (MemberID) REFERENCES  
Member(MemberID) ON DELETE CASCADE, FOREIGN KEY (BookID)  
REFERENCES Book(BookID) ON DELETE CASCADE );
```

```

Command Prompt - mysql -u x
+-----+
| BookID | int | NO | PRI | NULL |
| Title | varchar(100) | NO | | NULL |
| Author | varchar(60) | YES | | NULL |
| Price | decimal(8,2) | YES | | NULL |
| ISBN | varchar(20) | NO | UNI | NULL |
| Rating | decimal(2,1) | YES | | NULL |
| Stock | int | YES | | NULL |
| CategoryID | int | YES | MUL | NULL |
| PublisherID | int | YES | MUL | NULL |
+-----+
9 rows in set (0.04 sec)

mysql> DESCRIBE MEMBER;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| MemberID | int | NO | PRI | NULL |
| MemberName | varchar(60) | NO | | NULL |
| Age | int | YES | | NULL |
| Email | varchar(100) | NO | UNI | NULL |
| MembershipType | varchar(10) | YES | | NULL |
| ReferredBy | int | YES | MUL | NULL |
+-----+
6 rows in set (0.00 sec)

mysql> DESCRIBE PUBLISHER;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| PublisherID | int | NO | PRI | NULL |
| PublisherName | varchar(60) | NO | UNI | NULL |
| Country | varchar(40) | NO | | NULL |
+-----+
3 rows in set (0.00 sec)

mysql> |

```

● INSERT THE FULL DATASET

SYNTAX:

INSERT INTO table_name VALUES (value1, value2, ...);

QUERY:

INSERT INTO Category VALUES (1, 'Fiction');

INSERT INTO Category VALUES (2, 'Science');

INSERT INTO Category VALUES (3, 'History');

INSERT INTO Category VALUES (4, 'Cooking');

INSERT INTO Category VALUES (5, 'Technical');

INSERT INTO Publisher VALUES (1, 'Peacock House', 'India');

INSERT INTO Publisher VALUES (2, 'Northline Press', 'India');

INSERT INTO Publisher VALUES (3, 'Horizon Books', 'UK');

```
INSERT INTO Book VALUES (1, 'The Silent Peak','Rohan Iyer', 450,
'978-93-100-0001', 4.5, 12, 1, 1);

INSERT INTO Book VALUES (2, 'Modern Physics', 'Arjun Bose', 540,
'978-93-100-0002', 4.7, 5, 2, 2);

INSERT INTO Book VALUES (3, 'Bengal Chronicles','Sourav Ghosh', 380,
'978-93-100-0003', 4.1, 8, 3, 1);

INSERT INTO Book VALUES (4, 'Midnight in Kerala','Kavya Nair', 600,
'978-93-100-0004', NULL, 3, 1, 3);

INSERT INTO Book VALUES (5, 'Culinary Secrets','Priya Menon', 470,
'978-93-100-0005', 3.8, 6, 4, 2);

INSERT INTO Book VALUES (6, 'Data Structures Simplified','Meera Kulkarni', 620,
'978-93-100-0006', 4.2, 7, 5, 2);

INSERT INTO Book VALUES (7, 'Algorithms Unlocked','Meera Kulkarni', 580,
'978-93-100-0007', 4.6, 4, 5, 3);

INSERT INTO Book VALUES (8, 'Whispers of Rajasthan','Kavya Nair', 320,
'978-93-100-0008', NULL, 9, 1, 1);


INSERT INTO Member VALUES (501, 'Ananya Verma', 22, 'ananya@mail.com',
'Premium', NULL);

INSERT INTO Member VALUES (502, 'Rajesh Kumar', 35, 'rajesh@mail.com',
'Regular', 501);

INSERT INTO Member VALUES (503, 'Farah Sheikh', 27, 'farah@mail.com',
'Student', 501);

INSERT INTO Member VALUES (504, 'Kabir Singh', 30, 'kabir@mail.com',
'Regular', 502);
```

```
INSERT INTO Member VALUES (505, 'Ishaan Verma', 24, 'ishaan@mail.com',  
'Student', NULL);
```

```
INSERT INTO Member VALUES (506, 'Meera Joshi', 29, 'meera@mail.com',  
'Premium', 502);
```

```
INSERT INTO IssueRecord VALUES (501, 1, '2026-03-01', '2026-03-10', 0);
```

```
INSERT INTO IssueRecord VALUES (501, 6, '2026-03-05', NULL, 0);
```

```
INSERT INTO IssueRecord VALUES (502, 2, '2026-03-02', '2026-03-09', 0);
```

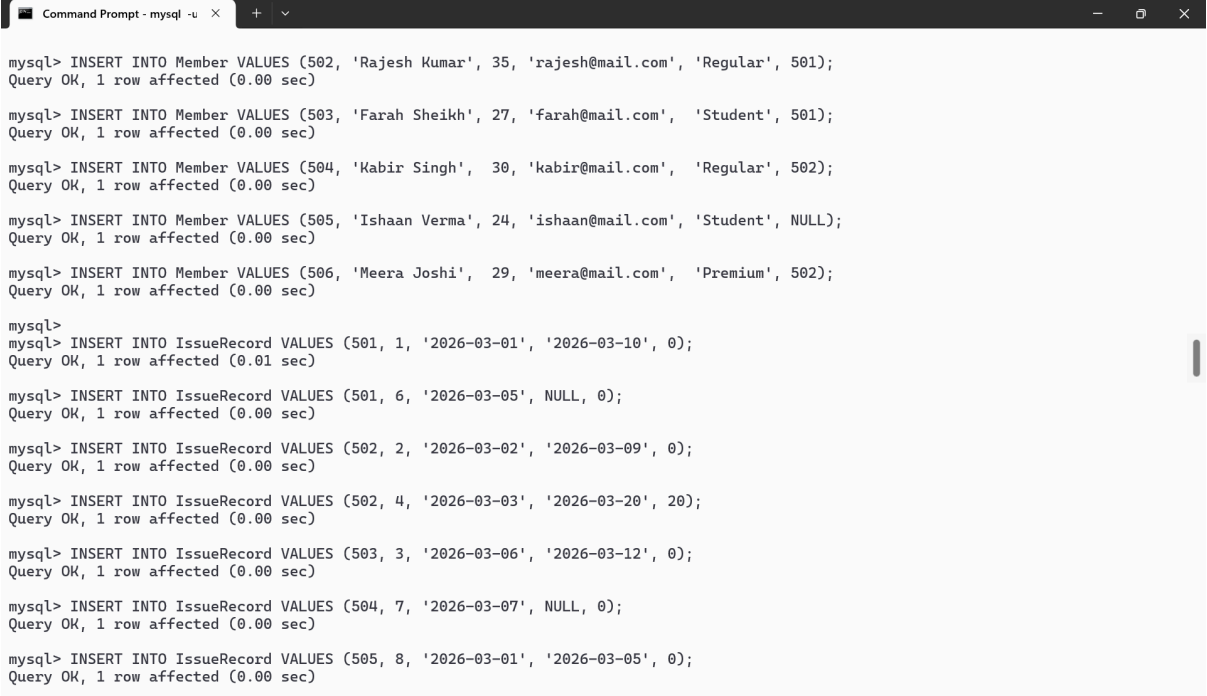
```
INSERT INTO IssueRecord VALUES (502, 4, '2026-03-03', '2026-03-20', 20);
```

```
INSERT INTO IssueRecord VALUES (503, 3, '2026-03-06', '2026-03-12', 0);
```

```
INSERT INTO IssueRecord VALUES (504, 7, '2026-03-07', NULL, 0);
```

```
INSERT INTO IssueRecord VALUES (505, 8, '2026-03-01', '2026-03-05', 0);
```

```
INSERT INTO IssueRecord VALUES (506, 5, '2026-03-04', '2026-03-11', 10);
```



```
Command Prompt - mysql -u x + v  
  
mysql> INSERT INTO Member VALUES (502, 'Rajesh Kumar', 35, 'rajesh@mail.com', 'Regular', 501);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Member VALUES (503, 'Farah Sheikh', 27, 'farah@mail.com', 'Student', 501);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Member VALUES (504, 'Kabir Singh', 30, 'kabir@mail.com', 'Regular', 502);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Member VALUES (505, 'Ishaan Verma', 24, 'ishaan@mail.com', 'Student', NULL);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO Member VALUES (506, 'Meera Joshi', 29, 'meera@mail.com', 'Premium', 502);  
Query OK, 1 row affected (0.00 sec)  
  
mysql>  
mysql> INSERT INTO IssueRecord VALUES (501, 1, '2026-03-01', '2026-03-10', 0);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (501, 6, '2026-03-05', NULL, 0);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (502, 2, '2026-03-02', '2026-03-09', 0);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (502, 4, '2026-03-03', '2026-03-20', 20);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (503, 3, '2026-03-06', '2026-03-12', 0);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (504, 7, '2026-03-07', NULL, 0);  
Query OK, 1 row affected (0.00 sec)  
  
mysql> INSERT INTO IssueRecord VALUES (505, 8, '2026-03-01', '2026-03-05', 0);  
Query OK, 1 row affected (0.00 sec)
```

● VIEW THE FULL DATASET

SYNTAX:

SELECT * FROM table_name;

QUERY:

SELECT * FROM IssueRecord;

SELECT * FROM Member;

SELECT * FROM Category;

SELECT * FROM Publisher;

SELECT * FROM book;

Command Prompt - mysql -u

BookID	Title	Author	Price	ISBN	Rating	Stock	CategoryID	PublisherID
1	The Silent Peak	Rohan Iyer	450.00	978-93-100-0001	4.5	12	1	1
2	Modern Physics	Arjun Bose	540.00	978-93-100-0002	4.7	5	2	2
3	Bengal Chronicles	Sourav Ghosh	380.00	978-93-100-0003	4.1	8	3	1
4	Midnight in Kerala	Kavya Nair	600.00	978-93-100-0004	NULL	3	1	3
5	Culinary Secrets	Priya Menon	470.00	978-93-100-0005	3.8	6	4	2
6	Data Structures Simplified	Meera Kulkarni	620.00	978-93-100-0006	4.2	7	5	2
7	Algorithms Unlocked	Meera Kulkarni	580.00	978-93-100-0007	4.6	4	5	3
8	Whispers of Rajasthan	Kavya Nair	320.00	978-93-100-0008	NULL	9	1	1

8 rows in set (0.01 sec)

mysql> select * from member;

MemberID	MemberName	Age	Email	MembershipType	ReferredBy
501	Ananya Verma	22	ananya@mail.com	Premium	NULL
502	Rajesh Kumar	35	rajesh@mail.com	Regular	501
503	Farah Sheikh	27	farah@mail.com	Student	501
504	Kabir Singh	30	kabir@mail.com	Regular	502
505	Ishaan Verma	24	ishaan@mail.com	Student	NULL
506	Meera Joshi	29	meera@mail.com	Premium	502

6 rows in set (0.00 sec)

mysql> select* from publisher;

PublisherID	PublisherName	Country
1	Peacock House	India
2	Northline Press	India
3	Horizon Books	UK

3 rows in set (0.00 sec)

mysql>

● CONSTRAINT VIOLATIONS

SYNTAX:

INSERT INTO table_name VALUES (value1, value2, ...); -- each one deliberately breaks one constraint

QUERY:

INSERT INTO Book VALUES (9, NULL, 'Someone', 300, '978-93-100-0009', NULL, 5, 1, 1);

```
INSERT INTO Book VALUES (9, 'Copy', 'Someone', 300, '978-93-100-0001', NULL,
5, 1, 1);
```

```
INSERT INTO Book VALUES (9, 'Free Book', 'Someone', -50, '978-93-100-0009',
NULL, 5, 1, 1);
```

```
INSERT INTO Book VALUES (9, 'Orphan Book', 'Someone', 300, '978-93-100-0009',
NULL, 5, 99, 1);
```

```
INSERT INTO Book VALUES (1, 'Dup Book', 'Someone', 300, '978-93-100-0010',
NULL, 5, 1, 1);
```

```
mysql> INSERT INTO Book VALUES (9, NULL, 'Someone', 300, '978-93-100-0009', NULL, 5, 1, 1);
ERROR 1048 (23000): Column 'Title' cannot be null
mysql> INSERT INTO Book VALUES (9, 'Copy', 'Someone', 300, '978-93-100-0001', NULL, 5, 1, 1);
ERROR 1062 (23000): Duplicate entry '978-93-100-0001' for key 'book.ISBN'
mysql> INSERT INTO Book VALUES (9, 'Free Book', 'Someone', -50, '978-93-100-0009', NULL, 5, 1, 1);
ERROR 3819 (HY000): Check constraint 'book_chk_1' is violated.
mysql> INSERT INTO Book VALUES (9, 'Orphan Book', 'Someone', 300, '978-93-100-0009', NULL, 5, 99, 1);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('libraryprojectdb`.`book`, CONSTRAINT `book_ibfk_1` FOREIGN KEY (`CategoryID`) REFERENCES `category` (`CategoryID`) ON DELETE CASCADE ON UPDATE CASCADE)
mysql> INSERT INTO Book VALUES (1, 'Dup Book', 'Someone', 300, '978-93-100-0010', NULL, 5, 1, 1);
ERROR 1062 (23000): Duplicate entry '1' for key 'book.PRIMARY'
mysql> |
```

● ALTER TABLE

SYNTAX:

```
ALTER TABLE table_name ADD COLUMN column_name datatype;
```

QUERY:

```
ALTER TABLE Member ADD COLUMN Phone VARCHAR(15);
```

```
DESCRIBE Member;
```

```
mysql> ALTER TABLE Member ADD COLUMN Phone VARCHAR(15);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> describe member;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| MemberID       | int           | NO   | PRI | NULL    |       |
| MemberName     | varchar(60)   | NO   |     | NULL    |       |
| Age            | int           | YES  |     | NULL    |       |
| Email          | varchar(100)  | NO   |     | NULL    |       |
| MembershipType | varchar(10)   | YES  |     | NULL    |       |
| ReferredBy     | int           | YES  |     | NULL    |       |
| Phone          | varchar(15)   | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.07 sec)

mysql> |
```

● DELETE DEMONSTRATING CASCADE

SYNTAX:

DELETE FROM table_name WHERE condition;

QUERY:

DELETE FROM Category WHERE CategoryID = 4;

SELECT * FROM Book;

SELECT * FROM IssueRecord;

```
mysql> DELETE FROM Category WHERE CategoryID = 4;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Book;
+----+-----+-----+-----+-----+-----+-----+-----+
| BookID | Title                | Author      | Price | ISBN          | Rating | Stock | CategoryID | PublisherID |
+----+-----+-----+-----+-----+-----+-----+-----+
| 1      | The Silent Peak      | Rohan Iyer  | 450.00 | 978-93-100-0001 | 4.5    | 12    | 1          | 1          |
| 2      | Modern Physics       | Arjun Bose  | 540.00 | 978-93-100-0002 | 4.7    | 5     | 2          | 2          |
| 3      | Bengal Chronicles    | Sourav Ghosh | 380.00 | 978-93-100-0003 | 4.1    | 8     | 3          | 1          |
| 4      | Midnight in Kerala   | Kavya Nair  | 600.00 | 978-93-100-0004 | NULL   | 3     | 1          | 3          |
| 6      | Data Structures Simplified | Meera Kulkarni | 620.00 | 978-93-100-0006 | 4.2    | 7     | 5          | 2          |
| 7      | Algorithms Unlocked  | Meera Kulkarni | 580.00 | 978-93-100-0007 | 4.6    | 4     | 5          | 3          |
| 8      | Whispers of Rajasthan | Kavya Nair  | 320.00 | 978-93-100-0008 | NULL   | 9     | 1          | 1          |
+----+-----+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT * FROM IssueRecord;
+----+-----+-----+-----+-----+
| MemberID | BookID | IssueDate | ReturnDate | Fine |
+----+-----+-----+-----+-----+
| 501      | 1      | 2026-03-01 | 2026-03-10 | 0.00 |
| 501      | 6      | 2026-03-05 | NULL       | 0.00 |
| 502      | 2      | 2026-03-02 | 2026-03-09 | 0.00 |
| 502      | 4      | 2026-03-03 | 2026-03-20 | 20.00 |
| 503      | 3      | 2026-03-06 | 2026-03-12 | 0.00 |
| 504      | 7      | 2026-03-07 | NULL       | 0.00 |
| 505      | 8      | 2026-03-01 | 2026-03-05 | 0.00 |
+----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> |
```

2. BASIC SQL RETRIEVAL

• DISTINCT

SYNTAX:

SELECT DISTINCT column1 FROM table_name;

SELECT DISTINCT column1, column2 FROM table_name WHERE condition;

QUERY:

SELECT DISTINCT CategoryID FROM Book;

SELECT DISTINCT Author, CategoryID FROM Book WHERE CategoryID = 1;

```
mysql> SELECT DISTINCT CategoryID FROM Book;
+-----+
| CategoryID |
+-----+
| 1          |
| 2          |
| 3          |
| 5          |
+-----+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT DISTINCT Author, CategoryID FROM Book WHERE CategoryID = 1;
+-----+-----+
| Author      | CategoryID |
+-----+-----+
| Rohan Iyer  | 1          |
| Kavya Nair  | 1          |
+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```

- **LIKE**

SYNTAX:

SELECT columns FROM table_name WHERE column LIKE 'pattern%';

SELECT columns FROM table_name WHERE column LIKE '%pattern%';

SELECT columns FROM table_name WHERE column LIKE '%pattern';

SELECT columns FROM table_name WHERE column LIKE '_pattern';

QUERY:

SELECT Title FROM Book WHERE Title LIKE 'M%';

SELECT Title FROM Book WHERE Title LIKE '%an%';

SELECT Title FROM Book WHERE Title LIKE '%s';

SELECT Title FROM Book WHERE Author LIKE '_eera Kulkarni';



```

Command Prompt - mysql -u <...>
+-----+
| Title |
+-----+
| Modern Physics |
| Midnight in Kerala |
+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT Title FROM Book WHERE Title LIKE 'M%';
+-----+
| Title |
+-----+
| Whispers of Rajasthan |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT Title FROM Book WHERE Title LIKE '%s';
+-----+
| Title |
+-----+
| Modern Physics |
| Bengal Chronicles |
+-----+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT Title FROM Book WHERE Author LIKE '_eera Kulkarni';
+-----+
| Title |
+-----+
| Data Structures Simplified |
| Algorithms Unlocked |
+-----+
2 rows in set (0.00 sec)

mysql> |

```

- **COMBINED QUERY (MULTIPLE OPERATORS TOGETHER)**

SYNTAX:

SELECT columns

FROM table_name

WHERE column_a BETWEEN low AND high

AND column_b IS NOT NULL

AND column_c < > value

ORDER BY column_a DESC;

QUERY:

SELECT Title, CategoryID, Price, Rating

FROM Book

WHERE Price BETWEEN 350 AND 600

AND Rating IS NOT NULL

AND CategoryID < > 1

ORDER BY Price DESC;

```
mysql> SELECT Title, CategoryID, Price, Rating
-> FROM Book
-> WHERE Price BETWEEN 350 AND 600
-> AND Rating IS NOT NULL
-> AND CategoryID <> 1
-> ORDER BY Price DESC;
```

Title	CategoryID	Price	Rating
Algorithms Unlocked	5	580.00	4.6
Modern Physics	2	540.00	4.7
Bengal Chronicles	3	380.00	4.1

```
3 rows in set (0.07 sec)

mysql> |
```

3. AGGREGATE FUNCTIONS

● COUNT()

SYNTAX:

SELECT COUNT(*) FROM table_name;

SELECT COUNT(DISTINCT column) FROM table_name;

QUERY:

SELECT COUNT(*) AS TotalBooks FROM Book;

SELECT COUNT(DISTINCT CategoryID) AS CategoriesInUse FROM Book;

```
mysql> SELECT COUNT(*) AS TotalBooks FROM Book;
```

TotalBooks
7

```
1 row in set (0.01 sec)

mysql> SELECT COUNT(DISTINCT CategoryID) AS CategoriesInUse FROM Book;
```

CategoriesInUse
4

```
1 row in set (0.00 sec)

mysql> |
```

- **AVG()**

SYNTAX:

SELECT AVG(column) FROM table_name;

QUERY:

SELECT AVG(Fine) AS AverageFine FROM IssueRecord;

```
mysql> SELECT SUM(Price) AS TotalTechnicalValue
-> FROM Book
-> WHERE CategoryID = 5;
+-----+
| TotalTechnicalValue |
+-----+
|          1200.00 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(Fine) AS AverageFine FROM IssueRecord;
+-----+
| AverageFine |
+-----+
|    2.857143 |
+-----+
1 row in set (0.01 sec)

mysql> |
```

- **MIN() AND MAX()**

SYNTAX:

SELECT MIN(column), MAX(column) FROM table_name;

QUERY:

SELECT MIN(Price) AS CheapestBook, MAX(Price) AS CostliestBook FROM
Book;

```
mysql> SELECT MIN(Price) AS CheapestBook, MAX(Price) AS CostliestBook FROM Book;
+-----+-----+
| CheapestBook | CostliestBook |
+-----+-----+
|          320.00 |          620.00 |
+-----+-----+
1 row in set (0.01 sec)

mysql> |
```

- **COMBINING WHERE, GROUP BY, AND HAVING**

SYNTAX:

SELECT column, AGGREGATE_FUNCTION(column)

FROM table_name

WHERE row_condition

GROUP BY column

HAVING aggregate_condition

ORDER BY column;

QUERY:

SELECT CategoryID, COUNT(*) AS PremiumBookCount

FROM Book

WHERE Price > 400

GROUP BY CategoryID

HAVING COUNT(*) > 1

ORDER BY CategoryID;

```
mysql> SELECT CategoryID, COUNT(*) AS PremiumBookCount
-> FROM Book
-> WHERE Price > 400
-> GROUP BY CategoryID
-> HAVING COUNT(*) > 1
-> ORDER BY CategoryID;
```

CategoryID	PremiumBookCount
1	2
5	2

```
2 rows in set (0.01 sec)

mysql> |
```

● NUMERIC FUNCTIONS

SYNTAX:

SELECT ROUND(n, d), ABS(n), CEIL(n), FLOOR(n), POWER(n, p), SQRT(n),

MOD(n, d), TRUNCATE(n, d);

QUERY:

SELECT BookID, ROUND(AVG(Fine), 2) AS AvgFine

FROM IssueRecord GROUP BY BookID;

```
mysql> SELECT BookID, ROUND(AVG(Fine), 2) AS AvgFine
-> FROM IssueRecord
-> GROUP BY BookID;
```

BookID	AvgFine
1	0.00
2	0.00
3	0.00
4	20.00
6	0.00
7	0.00
8	0.00

7 rows in set (0.01 sec)

● STRING FUNCTIONS

SYNTAX:

SELECT UPPER(s), LOWER(s), LENGTH(s), CONCAT(a, b), SUBSTRING(s,
start, len), TRIM(s), REPLACE(s, old, new), LEFT(s, n), RIGHT(s, n);

QUERY:

SELECT UPPER(Title) AS TitleUpper, LENGTH(Title) AS TitleLength

FROM Book

WHERE CategoryID = 5;

SELECT CONCAT(Author, ' - ', Title) AS Label FROM Book WHERE

CategoryID=1;

```
mysql> SELECT UPPER(Title) AS TitleUpper, LENGTH(Title) AS TitleLength
-> FROM Book
-> WHERE CategoryID = 5;
```

TitleUpper	TitleLength
DATA STRUCTURES SIMPLIFIED	26
ALGORITHMS UNLOCKED	19

2 rows in set (0.01 sec)

```
mysql>
mysql> SELECT CONCAT(Author, ' - ', Title) AS Label
-> FROM Book
-> WHERE CategoryID = 1;
```

Label
Rohan Iyer - The Silent Peak
Kavya Nair - Midnight in Kerala
Kavya Nair - Whispers of Rajasthan

3 rows in set (0.00 sec)

```
mysql> |
```

4. JOINS & NESTED QUERIES

● INNER JOIN

SYNTAX:

SELECT columns

FROM table1 T1

INNER JOIN table2 T2

ON T1.common_column = T2.common_column;

QUERY:

SELECT B.BookID, B.Title, C.CategoryName

FROM Book B

INNER JOIN Category C

ON B.CategoryID = C.CategoryID;

```
mysql> SELECT B.BookID, B.Title, C.CategoryName
-> FROM Book B
-> INNER JOIN Category C
-> ON B.CategoryID = C.CategoryID;
```

BookID	Title	CategoryName
1	The Silent Peak	Fiction
4	Midnight in Kerala	Fiction
8	Whispers of Rajasthan	Fiction
3	Bengal Chronicles	History
2	Modern Physics	Science
6	Data Structures Simplified	Technical
7	Algorithms Unlocked	Technical

```
7 rows in set (0.01 sec)

mysql> |
```

● LEFT JOIN (LEFT OUTER JOIN)

SYNTAX:

SELECT columns

FROM table1 T1

LEFT JOIN table2 T2

ON T1.common_column = T2.common_column;

QUERY:

```

SELECT B.Title, P.PublisherName

FROM Book B

LEFT JOIN Publisher P

ON B.PublisherID = P.PublisherID;

```

```

mysql> SELECT B.Title, P.PublisherName
-> FROM Book B
-> LEFT JOIN Publisher P
-> ON B.PublisherID = P.PublisherID;
+-----+-----+
| Title           | PublisherName |
+-----+-----+
| The Silent Peak  | Peacock House |
| Modern Physics  | Northline Press |
| Bengal Chronicles | Peacock House |
| Midnight in Kerala | NULL         |
| Data Structures Simplified | Northline Press |
| Algorithms Unlocked | NULL         |
| Whispers of Rajasthan | Peacock House |
+-----+-----+
7 rows in set (0.01 sec)

mysql> |

```

● RIGHT JOIN (RIGHT OUTER JOIN)

SYNTAX:

```

SELECT columns

FROM table1 T1

RIGHT JOIN table2 T2

ON T1.common_column = T2.common_column;

```

QUERY:

```

SELECT B.Title, C.CategoryName

FROM Book B

RIGHT JOIN Category C

ON B.CategoryID = C.CategoryID;

```

```

mysql> SELECT B.Title, C.CategoryName
-> FROM Book B
-> RIGHT JOIN Category C
-> ON B.CategoryID = C.CategoryID;
+-----+-----+
| Title           | CategoryName |
+-----+-----+
| The Silent Peak  | Fiction      |
| Midnight in Kerala | Fiction      |
| Whispers of Rajasthan | Fiction      |
| Bengal Chronicles | History      |
| Modern Physics  | Science      |
| Data Structures Simplified | Technical    |
| Algorithms Unlocked | Technical    |
+-----+-----+
7 rows in set (0.01 sec)

mysql> |

```

- **CORRELATED SUBQUERY**

SYNTAX:

SELECT columns

FROM table_name T1

WHERE column = (

 SELECT AGGREGATE_FUNCTION(column)

 FROM table_name T2

 WHERE T1.grouping_column = T2.grouping_column

);

QUERY:

SELECT Title, Price, CategoryID

FROM Book B1

WHERE Price = (

 SELECT MAX(Price)

 FROM Book B2

 WHERE B1.CategoryID = B2.CategoryID

);

```
mysql> SELECT Title, Price, CategoryID
-> FROM Book B1
-> WHERE Price = (
->     SELECT MAX(Price)
->     FROM Book B2
->     WHERE B1.CategoryID = B2.CategoryID
-> );
```

Title	Price	CategoryID
Modern Physics	540.00	2
Bengal Chronicles	380.00	3
Midnight in Kerala	600.00	1
Data Structures Simplified	620.00	5

4 rows in set (0.00 sec)

```
mysql> |
```

5. VIEWS & TRANSACTION CONTROL

● CREATE A VIEW

SYNTAX:

CREATE VIEW view_name AS

SELECT columns

FROM table_name

WHERE condition;

QUERY:

CREATE VIEW View_PremiumBooks AS

SELECT BookID, Title, Price, CategoryID

FROM Book

WHERE Price > 400;

SELECT * FROM View_PremiumBooks;

```
mysql> CREATE VIEW View_PremiumBooks AS
-> SELECT BookID, Title, Price, CategoryID
-> FROM Book
-> WHERE Price > 400;
Query OK, 0 rows affected (0.11 sec)

mysql>
mysql> SELECT * FROM View_PremiumBooks;
+-----+-----+-----+-----+
| BookID | Title                | Price  | CategoryID |
+-----+-----+-----+-----+
| 1      | The Silent Peak      | 450.00 | 1          |
| 2      | Modern Physics       | 540.00 | 2          |
| 4      | Midnight in Kerala   | 600.00 | 1          |
| 6      | Data Structures Simplified | 620.00 | 5          |
| 7      | Algorithms Unlocked  | 580.00 | 5          |
+-----+-----+-----+-----+
5 rows in set (0.02 sec)

mysql> |
```

● UPDATE THE BASE TABLE THROUGH A VIEW

SYNTAX:

UPDATE view_name SET column = value WHERE condition;

QUERY:

UPDATE View_PremiumBooks SET Price = 560 WHERE BookID = 7;

SELECT BookID, Title, Price FROM Book WHERE BookID = 7;

```
mysql> UPDATE View_PremiumBooks SET Price = 560 WHERE BookID = 7;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT BookID, Title, Price FROM Book WHERE BookID = 7;
+-----+-----+-----+
| BookID | Title           | Price |
+-----+-----+-----+
|      7 | Algorithms Unlocked | 560.00 |
+-----+-----+-----+
1 row in set (0.07 sec)

mysql> |
```

● DELETE (DROP) A VIEW

SYNTAX:

DROP VIEW view_name;

QUERY:

DROP VIEW View_PremiumBooks;

```
mysql> DROP VIEW View_PremiumBooks;
Query OK, 0 rows affected (0.06 sec)

mysql> |
```

● COMMIT

SYNTAX:

START TRANSACTION;

-- statements

COMMIT;

QUERY:

START TRANSACTION;

INSERT INTO Book VALUES (9, 'Poems of the Coast', 'Ishaan Verma', 340,

'978-93-100-0009', NULL, 10, 1, 1);

COMMIT;

SELECT * FROM Book WHERE BookID = 9;

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Book VALUES (9, 'Poems of the Coast', 'Ishaan Verma', 340, '978-93-100-0009', NULL, 10, 1, 1);
ERROR 1062 (23000): Duplicate entry '9' for key 'book.PRIMARY'
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Book WHERE BookID = 9;
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| BookID | Title           | Author   | Price | ISBN           | Rating | Stock | CategoryID | PublisherID |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| 9      | Poems of the Coast | Ishaan Verma | 340.00 | 978-93-100-0009 | NULL   | 10    | 1          | 1          |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

● ROLLBACK

SYNTAX:

START TRANSACTION;

-- statements

ROLLBACK;

QUERY:

START TRANSACTION;

DELETE FROM Book WHERE BookID = 3;

SELECT * FROM Book; -- Bengal Chronicles missing here

ROLLBACK;

SELECT * FROM Book; -- Bengal Chronicles restored

```
Command Prompt - mysql -u <...>
mysql> DELETE FROM Book WHERE BookID = 3;
Query OK, 1 row affected (0.04 sec)

mysql> SELECT * FROM Book; -- Bengal Chronicles missing here
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| BookID | Title           | Author   | Price | ISBN           | Rating | Stock | CategoryID | PublisherID |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1      | The Silent Peak  | Rohan Iyer | 450.00 | 978-93-100-0001 | 4.5    | 12    | 1          | 1          |
| 2      | Modern Physics  | Arjun Bose | 540.00 | 978-93-100-0002 | 4.7    | 5     | 2          | 2          |
| 4      | Midnight in Kerala | Kavya Nair | 600.00 | 978-93-100-0004 | NULL   | 3     | 1          | NULL       |
| 6      | Data Structures Simplified | Meera Kulkarni | 620.00 | 978-93-100-0006 | 4.2    | 7     | 5          | 2          |
| 7      | Algorithms Unlocked | Meera Kulkarni | 560.00 | 978-93-100-0007 | 4.6    | 4     | 5          | NULL       |
| 8      | Whispers of Rajasthan | Kavya Nair | 320.00 | 978-93-100-0008 | NULL   | 9     | 1          | 1          |
| 9      | Poems of the Coast | Ishaan Verma | 340.00 | 978-93-100-0009 | NULL   | 10    | 1          | 1          |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.04 sec)

mysql> SELECT * FROM Book; -- Bengal Chronicles restored
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| BookID | Title           | Author   | Price | ISBN           | Rating | Stock | CategoryID | PublisherID |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1      | The Silent Peak  | Rohan Iyer | 450.00 | 978-93-100-0001 | 4.5    | 12    | 1          | 1          |
| 2      | Modern Physics  | Arjun Bose | 540.00 | 978-93-100-0002 | 4.7    | 5     | 2          | 2          |
| 3      | Bengal Chronicles | Sourav Ghosh | 380.00 | 978-93-100-0003 | 4.1    | 8     | 3          | 1          |
| 4      | Midnight in Kerala | Kavya Nair | 600.00 | 978-93-100-0004 | NULL   | 3     | 1          | NULL       |
| 6      | Data Structures Simplified | Meera Kulkarni | 620.00 | 978-93-100-0006 | 4.2    | 7     | 5          | 2          |
| 7      | Algorithms Unlocked | Meera Kulkarni | 560.00 | 978-93-100-0007 | 4.6    | 4     | 5          | NULL       |
| 8      | Whispers of Rajasthan | Kavya Nair | 320.00 | 978-93-100-0008 | NULL   | 9     | 1          | 1          |
| 9      | Poems of the Coast | Ishaan Verma | 340.00 | 978-93-100-0009 | NULL   | 10    | 1          | 1          |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> |
```

- **SAVEPOINT (PARTIAL ROLLBACK)**

SYNTAX:

```
START TRANSACTION;

    -- statements

SAVEPOINT savepoint_name;

    -- more statements

ROLLBACK TO savepoint_name;

COMMIT;

START TRANSACTION;

    -- statements

ROLLBACK;
```

QUERY:

```
START TRANSACTION;

UPDATE Book SET Price = Price + 20 WHERE CategoryID = 1;

SAVEPOINT sp1;

UPDATE Book SET Price = Price + 30 WHERE CategoryID = 5;

SAVEPOINT sp2;

DELETE FROM Book WHERE BookID = 8;

ROLLBACK TO sp2;

COMMIT;

SELECT * FROM Book ORDER BY BookID;
```

```

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Book SET Price = Price + 30 WHERE CategoryID = 5;
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> SAVEPOINT sp2;
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM Book WHERE BookID = 8;
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK TO sp2;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> SELECT * FROM Book ORDER BY BookID;
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| BookID | Title                | Author    | Price | ISBN          | Rating | Stock | CategoryID | PublisherID |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1      | The Silent Peak      | Rohan Iyer | 470.00 | 978-93-100-0001 | 4.5    | 12    | 1          | 1          |
| 2      | Modern Physics      | Arjun Bose | 540.00 | 978-93-100-0002 | 4.7    | 5     | 2          | 2          |
| 3      | Bengal Chronicles    | Sourav Ghosh | 380.00 | 978-93-100-0003 | 4.1    | 8     | 3          | 1          |
| 4      | Midnight in Kerala   | Kavya Nair | 620.00 | 978-93-100-0004 | NULL   | 3     | 1          | NULL       |
| 6      | Data Structures Simplified | Meera Kulkarni | 650.00 | 978-93-100-0006 | 4.2    | 7     | 5          | 2          |
| 7      | Algorithms Unlocked  | Meera Kulkarni | 590.00 | 978-93-100-0007 | 4.6    | 4     | 5          | NULL       |
| 8      | Whispers of Rajasthan | Kavya Nair | 340.00 | 978-93-100-0008 | NULL   | 9     | 1          | 1          |
| 9      | Poems of the Coast   | Ishaan Verma | 360.00 | 978-93-100-0009 | NULL   | 10    | 1          | 1          |
+----+-----+-----+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql>

```

6. DATABASE SECURITY & USER PRIVILEGES

● CREATE DATABASE USERS

SYNTAX:

CREATE USER user_name IDENTIFIED BY 'password';

QUERY:

CREATE USER librarian_user IDENTIFIED BY 'lib@12345';

CREATE USER member_user IDENTIFIED BY 'mem@123';

```

mysql> CREATE USER librarian_user IDENTIFIED BY 'lib@12345';
Query OK, 0 rows affected (0.12 sec)

mysql> CREATE USER member_user IDENTIFIED BY 'mem@123';
Query OK, 0 rows affected (0.05 sec)

mysql>

```

● GRANT PRIVILEGES

SYNTAX:

GRANT privilege ON object_name TO user_name;

GRANT privilege1, privilege2 ON object_name TO user_name;

GRANT ALL PRIVILEGES ON object_name TO user_name WITH GRANT
OPTION;

QUERY:

GRANT SELECT ON Book TO member_user;

GRANT SELECT, INSERT, UPDATE ON Book TO librarian_user;

GRANT ALL PRIVILEGES ON Book TO librarian_user WITH GRANT OPTION;

```
mysql> CREATE USER librarian_user IDENTIFIED BY 'lib@12345';  
Query OK, 0 rows affected (0.12 sec)  
  
mysql> CREATE USER member_user IDENTIFIED BY 'mem@123';  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> GRANT SELECT ON Book TO member_user;  
Query OK, 0 rows affected (0.04 sec)  
  
mysql>  
mysql> GRANT SELECT, INSERT, UPDATE ON Book TO librarian_user;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql>  
mysql> GRANT ALL PRIVILEGES ON Book TO librarian_user WITH GRANT OPTION;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> |
```

● VERIFY GRANTED PRIVILEGES

SYNTAX:

SELECT grantee, table_name, privilege_type

FROM information_schema.table_privileges

WHERE table_name = 'table_name';

QUERY:

SELECT grantee, table_name, privilege_type

FROM information_schema.table_privileges

WHERE table_name = 'book';

```
mysql> SELECT grantee, table_name, privilege_type
-> FROM information_schema.table_privileges
-> WHERE table_name = 'book';
```

grantee	table_name	privilege_type
'member_user'@'%'	book	SELECT
'librarian_user'@'%'	book	SELECT
'librarian_user'@'%'	book	INSERT
'librarian_user'@'%'	book	UPDATE
'librarian_user'@'%'	book	DELETE
'librarian_user'@'%'	book	CREATE
'librarian_user'@'%'	book	DROP
'librarian_user'@'%'	book	REFERENCES
'librarian_user'@'%'	book	INDEX
'librarian_user'@'%'	book	ALTER
'librarian_user'@'%'	book	CREATE VIEW
'librarian_user'@'%'	book	SHOW VIEW
'librarian_user'@'%'	book	TRIGGER

```
13 rows in set (0.01 sec)

mysql> |
```

● DEMONSTRATE ACCESS CONTROL (ALLOWED VS DENIED)

SYNTAX:

SELECT * FROM table_name;

DELETE FROM table_name WHERE condition;

QUERY:

SELECT * FROM Book;

DELETE FROM Book WHERE BookID = 3;

```
mysql> SELECT * FROM Book;
```

BookID	Title	Author	Price	ISBN	Rating	Stock	CategoryID	PublisherID
1	The Silent Peak	Rohan Iyer	470.00	978-93-100-0001	4.5	12	1	1
2	Modern Physics	Arjun Bose	540.00	978-93-100-0002	4.7	5	2	2
3	Bengal Chronicles	Sourav Ghosh	380.00	978-93-100-0003	4.1	8	3	1
4	Midnight in Kerala	Kavya Nair	620.00	978-93-100-0004	NULL	3	1	NULL
6	Data Structures Simplified	Meera Kulkarni	650.00	978-93-100-0006	4.2	7	5	2
7	Algorithms Unlocked	Meera Kulkarni	590.00	978-93-100-0007	4.6	4	5	NULL
8	Whispers of Rajasthan	Kavya Nair	340.00	978-93-100-0008	NULL	9	1	1
9	Poems of the Coast	Ishaan Verma	360.00	978-93-100-0009	NULL	10	1	1

```
8 rows in set (0.00 sec)

mysql>
mysql> DELETE FROM Book WHERE BookID = 3;
Query OK, 1 row affected (0.04 sec)

mysql> |
```

● REVOKE OPERATIONS

SYNTAX:

REVOKE privilege ON object_name FROM user_name;

REVOKE ALL PRIVILEGES ON object_name FROM user_name;

QUERY:

REVOKE INSERT ON Book FROM librarian_user;

REVOKE ALL PRIVILEGES ON Book FROM member_user;

SELECT grantee, table_name, privilege_type

FROM information_schema.table_privileges

WHERE table_name = 'book';

```
mysql> REVOKE INSERT ON Book FROM librarian_user;
Query OK, 0 rows affected (0.10 sec)

mysql> REVOKE ALL PRIVILEGES ON Book FROM member_user;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT grantee, table_name, privilege_type
-> FROM information_schema.table_privileges
-> WHERE table_name = 'book';
+-----+-----+-----+
| grantee          | table_name | privilege_type |
+-----+-----+-----+
| 'librarian_user'@'%' | book      | SELECT         |
| 'librarian_user'@'%' | book      | UPDATE         |
| 'librarian_user'@'%' | book      | DELETE         |
| 'librarian_user'@'%' | book      | CREATE         |
| 'librarian_user'@'%' | book      | DROP           |
| 'librarian_user'@'%' | book      | REFERENCES     |
| 'librarian_user'@'%' | book      | INDEX          |
| 'librarian_user'@'%' | book      | ALTER          |
| 'librarian_user'@'%' | book      | CREATE VIEW    |
| 'librarian_user'@'%' | book      | SHOW VIEW     |
| 'librarian_user'@'%' | book      | TRIGGER        |
+-----+-----+-----+
11 rows in set (0.00 sec)

mysql> |
```

● BACKUP AND RESTORE

SYNTAX:

CREATE TABLE table_name (...);

pg_dump -U username -d database_name -F c -f backup_file.dump

DROP TABLE table_name;

pg_restore -U username -d database_name -c backup_file.dump

QUERY:

CREATE TABLE Feedback (

FeedbackID INT PRIMARY KEY,

MemberID INT,

```

Comments VARCHAR(200),

Rating INT,

FOREIGN KEY (MemberID) REFERENCES Member(MemberID)

);

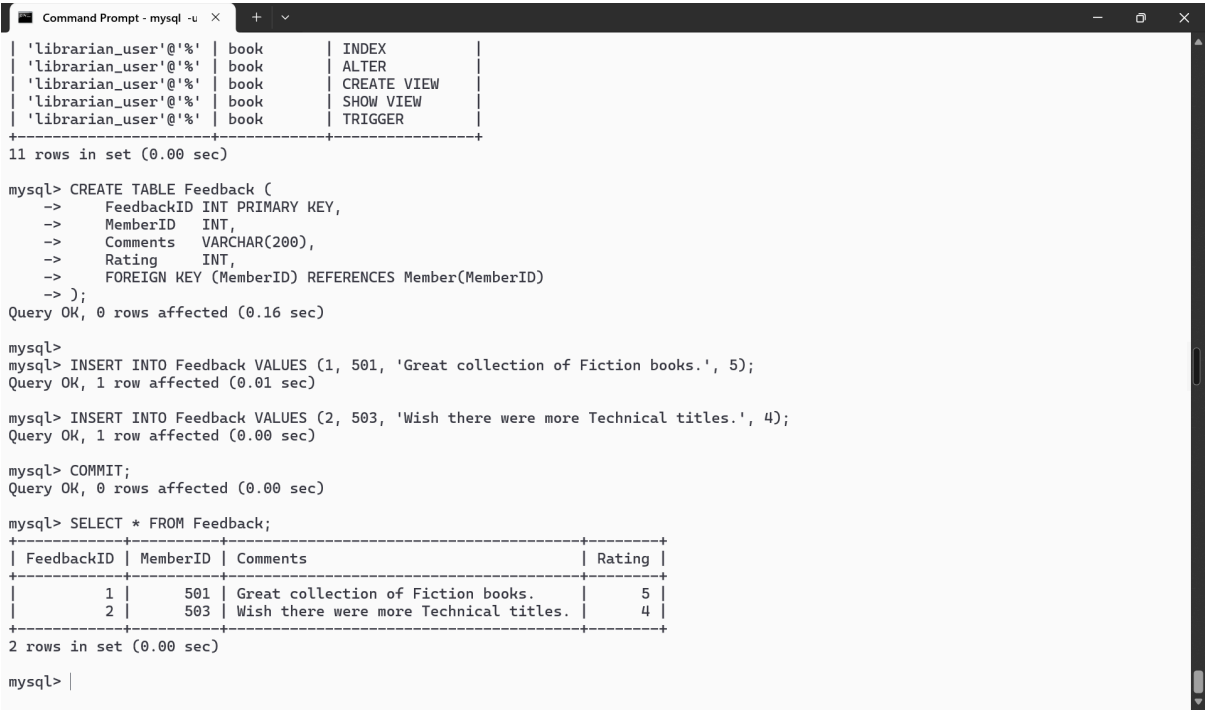
INSERT INTO Feedback VALUES (1, 501, 'Great collection of Fiction books.', 5);

INSERT INTO Feedback VALUES (2, 503, 'Wish there were more Technical titles.',

4);

COMMIT;

```



```

Command Prompt - mysql -u x
+ - v
| 'librarian_user'@'%' | book | INDEX |
| 'librarian_user'@'%' | book | ALTER |
| 'librarian_user'@'%' | book | CREATE VIEW |
| 'librarian_user'@'%' | book | SHOW VIEW |
| 'librarian_user'@'%' | book | TRIGGER |
+-----+-----+-----+
11 rows in set (0.00 sec)

mysql> CREATE TABLE Feedback (
-> FeedbackID INT PRIMARY KEY,
-> MemberID INT,
-> Comments VARCHAR(200),
-> Rating INT,
-> FOREIGN KEY (MemberID) REFERENCES Member(MemberID)
-> );
Query OK, 0 rows affected (0.16 sec)

mysql>
mysql> INSERT INTO Feedback VALUES (1, 501, 'Great collection of Fiction books.', 5);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Feedback VALUES (2, 503, 'Wish there were more Technical titles.', 4);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Feedback;
+-----+-----+-----+-----+
| FeedbackID | MemberID | Comments | Rating |
+-----+-----+-----+-----+
| 1 | 501 | Great collection of Fiction books. | 5 |
| 2 | 503 | Wish there were more Technical titles. | 4 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |

```

Command — take the backup (run in PowerShell/Command Prompt, not the MySQL prompt):

```
mysqldump -u root -p LibraryProjectDB > libraryproject_backup.sql
```

Query — simulate data loss:

```
DROP TABLE Feedback;
```

Command — restore (run in PowerShell, not SQL prompt):

```
mysql -u root -p LibraryProjectDB < libraryproject_backup.sql
```

Query — verify restoration

SELECT * FROM Feedback;

```

Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Feedback VALUES (2, 503, 'Wish there were more Technical titles.', 4);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM Feedback;
+-----+-----+-----+-----+
| FeedbackID | MemberID | Comments | Rating |
+-----+-----+-----+-----+
| 1 | 501 | Great collection of Fiction books. | 5 |
| 2 | 503 | Wish there were more Technical titles. | 4 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> pg_dump -U postgres -d LibraryProjectDB -F c -f libraryproject_backup.dump
-> pg_dump -U postgres -d LibraryProjectDB -F c -f libraryproject_backup.dump
-> mysqldump -u root -p LibraryProjectDB > libraryproject_backup.sql
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'pg_dump -U postgres -d LibraryProjectDB -F c -f libraryproject_backup.dump
pg_du' at line 1
mysql> DROP TABLE Feedback;
Query OK, 0 rows affected (0.08 sec)

mysql> SELECT * FROM Feedback;
+-----+-----+-----+-----+
| FeedbackID | MemberID | Comments | Rating |
+-----+-----+-----+-----+
| 1 | 501 | Great collection of Fiction books. | 5 |
| 2 | 503 | Wish there were more Technical titles. | 4 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |

```

7. STORED PROGRAM FUNDAMENTALS (MYSQL)

● PROGRAM 1: VARIABLES AND OPERATOR

SYNTAX:

DELIMITER //

CREATE PROCEDURE procedure_name()

BEGIN

DECLARE var1 INT DEFAULT value1;

DECLARE var2 INT DEFAULT value2;

DECLARE result INT;

SET result = var1 + var2; -- or -, *, /

SELECT result AS label;

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE PROCEDURE prog_Variables()

BEGIN

DECLARE a INT DEFAULT 18;

DECLARE b INT DEFAULT 5;

DECLARE sum_val, diff_val, prod_val INT;

DECLARE quot_val DECIMAL(6,2);

SET sum_val = a + b;

SET diff_val = a - b;

SET prod_val = a * b;

SET quot_val = a / b;

SELECT a AS A, b AS B, sum_val AS Sum, diff_val AS Difference,

prod_val AS Product, quot_val AS Quotient;

END //

DELIMITER ;

CALL prog_Variables();

```

mysql> DELIMITER //
mysql> CREATE PROCEDURE prog_Variables()
-> BEGIN
->     DECLARE a INT DEFAULT 18;
->     DECLARE b INT DEFAULT 5;
->     DECLARE sum_val, diff_val, prod_val INT;
->     DECLARE quot_val DECIMAL(6,2);
->
->     SET sum_val = a + b;
->     SET diff_val = a - b;
->     SET prod_val = a * b;
->     SET quot_val = a / b;
->
->     SELECT a AS A, b AS B, sum_val AS Sum, diff_val AS Difference,
->           prod_val AS Product, quot_val AS Quotient;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL prog_Variables();
+-----+-----+-----+-----+-----+-----+
| A | B | Sum | Difference | Product | Quotient |
+-----+-----+-----+-----+-----+-----+
| 18 | 5 | 23 | 13 | 90 | 3.60 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> |

```

● PROGRAM 2: CONDITIONAL STATEMENTS (IF-ELSEIF)

SYNTAX:

DELIMITER //

CREATE PROCEDURE procedure_name(IN p_id INT)

BEGIN

 DECLARE v_col datatype;

 SELECT column INTO v_col FROM table_name WHERE pk = p_id;

 IF v_col >= threshold1 THEN

 SELECT 'label1' AS Grade;

 ELSEIF v_col >= threshold2 THEN

 SELECT 'label2' AS Grade;

 ELSE

 SELECT 'label3' AS Grade;

 END IF;

END //

DELIMITER ;

QUERY:

```
DELIMITER //
```

```
CREATE PROCEDURE prog_PriceGrade(IN p_bookid INT)
```

```
BEGIN
```

```
    DECLARE v_title VARCHAR(100);
```

```
    DECLARE v_price DECIMAL(8,2);
```

```
    DECLARE v_grade VARCHAR(20);
```

```
    SELECT Title, Price INTO v_title, v_price FROM Book WHERE BookID = p_bookid;
```

```
    IF v_price >= 500 THEN
```

```
        SET v_grade = 'A - Premium';
```

```
    ELSEIF v_price >= 350 THEN
```

```
        SET v_grade = 'B - Standard';
```

```
    ELSE
```

```
        SET v_grade = 'C - Economy';
```

```
    END IF;
```

```
    SELECT v_title AS Book, v_price AS Price, v_grade AS Grade;
```

```
END //
```

```
DELIMITER ;
```

```
CALL prog_PriceGrade(3);
```

```
mysql> CALL prog_PriceGrade(3);
```

Book	Price	Grade
Bengal Chronicles	380.00	B - Standard

```
1 row in set (0.04 sec)
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> |
```

- **PROGRAM 3: CURSOR-BASED LOOP (DATABASE-RELATED PROBLEM)**

SYNTAX:

```
DELIMITER //
```

```
CREATE PROCEDURE procedure_name()
```

```
BEGIN
```

```
    DECLARE done INT DEFAULT FALSE;
```

```
    DECLARE v_col1 datatype;
```

```
    DECLARE v_col2 datatype;
```

```
    DECLARE cur CURSOR FOR SELECT col1, col2 FROM table_name;
```

```
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
```

```
    OPEN cur;
```

```
    read_loop: LOOP
```

```
        FETCH cur INTO v_col1, v_col2;
```

```
        IF done THEN
```

```
            LEAVE read_loop;
```

```
        END IF;
```

```
        -- process v_col1, v_col2
```

```
    END LOOP;
```

```
    CLOSE cur;
```

```
END //
```

```
DELIMITER ;
```

QUERY:

```
DELIMITER //

CREATE PROCEDURE prog_FeaturedBooks()

BEGIN

    DECLARE done INT DEFAULT FALSE;

    DECLARE v_title VARCHAR(100);

    DECLARE v_price DECIMAL(8,2);

    DECLARE v_avg_price DECIMAL(8,2);

    DECLARE cur CURSOR FOR SELECT Title, Price FROM Book;

    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

    DROP TEMPORARY TABLE IF EXISTS Results;

    CREATE TEMPORARY TABLE Results (Title VARCHAR(100), Price
DECIMAL(8,2), Status VARCHAR(20));

    SELECT AVG(Price) INTO v_avg_price FROM Book;


    OPEN cur;

    read_loop: LOOP

        FETCH cur INTO v_title, v_price;

        IF done THEN

            LEAVE read_loop;

        END IF;

        IF v_price > v_avg_price THEN

            INSERT INTO Results VALUES (v_title, v_price, 'Featured Pick');

        ELSE

            INSERT INTO Results VALUES (v_title, v_price, 'Standard Listing');

        END IF;

    END IF;
```

```

END LOOP;

CLOSE cur;

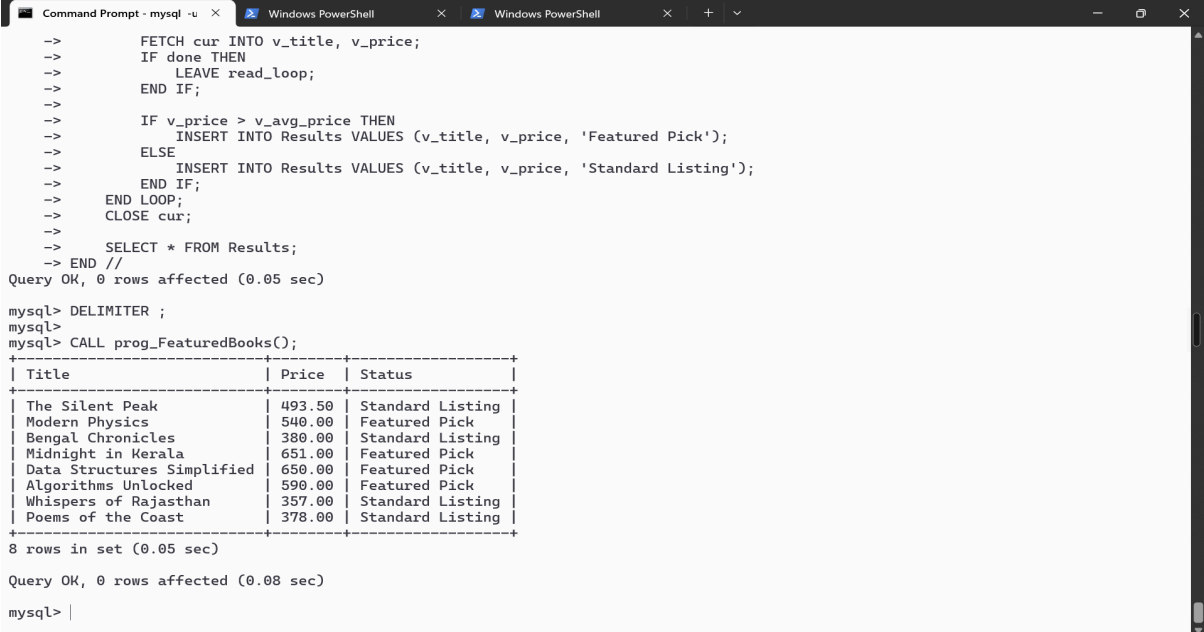
SELECT * FROM Results;

END //

DELIMITER ;

CALL prog_FeaturedBooks();

```



```

->      FETCH cur INTO v_title, v_price;
->      IF done THEN
->          LEAVE read_loop;
->      END IF;
->
->      IF v_price > v_avg_price THEN
->          INSERT INTO Results VALUES (v_title, v_price, 'Featured Pick');
->      ELSE
->          INSERT INTO Results VALUES (v_title, v_price, 'Standard Listing');
->      END IF;
->      END LOOP;
->      CLOSE cur;
->
->      SELECT * FROM Results;
->  END //
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL prog_FeaturedBooks();
+-----+-----+-----+
| Title                                     | Price | Status      |
+-----+-----+-----+
| The Silent Peak                         | 493.50 | Standard Listing |
| Modern Physics                         | 540.00 | Featured Pick   |
| Bengal Chronicles                       | 380.00 | Standard Listing |
| Midnight in Kerala                     | 651.00 | Featured Pick   |
| Data Structures Simplified              | 650.00 | Featured Pick   |
| Algorithms Unlocked                    | 590.00 | Featured Pick   |
| Whispers of Rajasthan                  | 357.00 | Standard Listing |
| Poems of the Coast                     | 378.00 | Standard Listing |
+-----+-----+-----+
8 rows in set (0.05 sec)

Query OK, 0 rows affected (0.08 sec)

mysql> |

```

8. STORED PROGRAM DATABASE OPERATIONS

● PROGRAM 1: INSERT WITH DUPLICATE-KEY HANDLING

SYNTAX:

```
DELIMITER //
```

```
CREATE PROCEDURE procedure_name()
```

```
BEGIN
```

```
    DECLARE EXIT HANDLER FOR 1062
```

```
    BEGIN
```

```
        SELECT 'Error: duplicate key.' AS Message;
```

```
    END;
```

```
INSERT INTO table_name VALUES (...);

SELECT 'Row inserted successfully.' AS Message;

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE PROCEDURE prog_InsertBook(

    IN p_id INT, IN p_title VARCHAR(100), IN p_author VARCHAR(60),

    IN p_price DECIMAL(8,2), IN p_isbn VARCHAR(20), IN p_stock INT,

    IN p_catid INT, IN p_pubid INT)

BEGIN

    DECLARE EXIT HANDLER FOR 1062

    BEGIN

        SELECT 'Error: BookID or ISBN already exists.' AS Message;

    END;

    INSERT INTO Book VALUES (p_id, p_title, p_author, p_price, p_isbn, NULL,

p_stock, p_catid, p_pubid);

    SELECT 'Book inserted successfully.' AS Message;

END //

DELIMITER ;

CALL prog_InsertBook(10, 'The Lighthouse Keeper', 'Devika Rao', 410.00,

'978-93-100-0010', 6, 2, 1);

CALL prog_InsertBook(10, 'Duplicate Entry', 'Someone', 300.00, '978-93-100-0011',

2, 1, 1);
```

```

Command Prompt - mysql -u x  Windows PowerShell  Windows PowerShell  x + v
-> IN p_id INT, IN p_title VARCHAR(100), IN p_author VARCHAR(60),
-> IN p_price DECIMAL(8,2), IN p_isbn VARCHAR(20), IN p_stock INT,
-> IN p_catid INT, IN p_pubid INT)
-> BEGIN
-> DECLARE EXIT HANDLER FOR 1062
-> BEGIN
-> SELECT 'Error: BookID or ISBN already exists.' AS Message;
-> END;
-> INSERT INTO Book VALUES (p_id, p_title, p_author, p_price, p_isbn, NULL, p_stock, p_catid, p_pubid);
-> SELECT 'Book inserted successfully.' AS Message;
-> END //
Query OK, 0 rows affected (0.07 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL prog_InsertBook(10, 'The Lighthouse Keeper', 'Devika Rao', 410.00, '978-93-100-0010', 6, 2, 1);
+-----+
| Message |
+-----+
| Book inserted successfully. |
+-----+
1 row in set (0.06 sec)

Query OK, 0 rows affected (0.06 sec)

mysql> CALL prog_InsertBook(10, 'Duplicate Entry', 'Someone', 300.00, '978-93-100-0011', 2, 1, 1);
+-----+
| Message |
+-----+
| Error: BookID or ISBN already exists. |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> |

```

9. STORED PROCEDURES AND FUNCTIONS

- **PROCEDURE: ISSUEBOOK (BUSINESS RULE — STOCK CAN'T GO NEGATIVE)**

SYNTAX:

DELIMITER //

CREATE PROCEDURE procedure_name(IN p_id INT)

BEGIN

DECLARE v_stock INT;

SELECT column INTO v_stock FROM table_name WHERE pk = p_id;

IF v_stock > 0 THEN

UPDATE table_name SET column = column - 1 WHERE pk = p_id;

ELSE

SELECT 'message' AS Result;

END IF;

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE PROCEDURE IssueBook(IN p_bookid INT)

BEGIN

 DECLARE v_stock INT;

 DECLARE v_title VARCHAR(100);

 SELECT Title, Stock INTO v_title, v_stock FROM Book WHERE BookID =
p_bookid;

 IF v_stock > 0 THEN

 UPDATE Book SET Stock = Stock - 1 WHERE BookID = p_bookid;

 SELECT CONCAT(v_title, ' issued. Remaining stock = ', v_stock - 1) AS

Result;

 ELSE

 SELECT CONCAT(v_title, ' is out of stock — cannot issue.') AS Result;

 END IF;

END //

DELIMITER ;

CALL IssueBook(4);

CALL IssueBook(4);

CALL IssueBook(4);

CALL IssueBook(4);

```
mysql> CALL IssueBook(4);
+-----+
| Result |
+-----+
| Midnight in Kerala issued. Remaining stock = 1 |
+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL IssueBook(4);
+-----+
| Result |
+-----+
| Midnight in Kerala issued. Remaining stock = 0 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL IssueBook(4);
+-----+
| Result |
+-----+
| Midnight in Kerala is out of stock ? cannot issue. |
+-----+
```

● FUNCTION: CALCULATEDISCOUNTEDPRICE

SYNTAX:

DELIMITER //

CREATE FUNCTION function_name(p1 datatype, p2 datatype)

RETURNS return_datatype

DETERMINISTIC

READS SQL DATA

BEGIN

 DECLARE v_result return_datatype;

 -- compute v_result

 RETURN v_result;

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE FUNCTION CalculateDiscountedPrice(p_bookid INT, p_discount_percent
DECIMAL(5,2))

RETURNS DECIMAL(8,2)

DETERMINISTIC

READS SQL DATA

BEGIN

DECLARE v_price DECIMAL(8,2);

DECLARE v_result DECIMAL(8,2);

SELECT Price INTO v_price FROM Book WHERE BookID = p_bookid;

SET v_result = v_price - (v_price * p_discount_percent / 100);

RETURN ROUND(v_result, 2);

END //

DELIMITER ;

SELECT Title, Price, CalculateDiscountedPrice(BookID, 15) AS DiscountedPrice

FROM Book

WHERE CategoryID = 1;

```

Command Prompt - mysql -u x  Windows PowerShell  Windows PowerShell
| Midnight in Kerala is out of stock ? cannot issue. |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION CalculateDiscountedPrice(p_bookid INT, p_discount_percent DECIMAL(5,2))
-> RETURNS DECIMAL(8,2)
-> DETERMINISTIC
-> READS SQL DATA
-> BEGIN
->     DECLARE v_price DECIMAL(8,2);
->     DECLARE v_result DECIMAL(8,2);
->
->     SELECT Price INTO v_price FROM Book WHERE BookID = p_bookid;
->     SET v_result = v_price - (v_price * p_discount_percent / 100);
->
->     RETURN ROUND(v_result, 2);
-> END //
Query OK, 0 rows affected (0.06 sec)

mysql> DELIMITER ;
mysql> SELECT Title, Price, CalculateDiscountedPrice(BookID, 15) AS DiscountedPrice
-> FROM Book
-> WHERE CategoryID = 1;
+-----+
| Title          | Price | DiscountedPrice |
+-----+
| The Silent Peak | 493.50 | 419.48 |
| Midnight in Kerala | 651.00 | 553.35 |
| Whispers of Rajasthan | 357.00 | 303.45 |
| Poems of the Coast | 378.00 | 321.30 |
+-----+
4 rows in set (0.01 sec)

mysql>

```

10. DATABASE TRIGGERS

- **CREATE THE AUDIT TABLE**

SYNTAX:

```
CREATE TABLE table_name (
    column1 datatype constraint,);
```

QUERY:

```
CREATE TABLE PriceAuditLog (
    LogID INT PRIMARY KEY AUTO_INCREMENT,
    BookID INT,
    OldPrice DECIMAL(8,2),
    NewPrice DECIMAL(8,2),
    ChangedOn DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

```
mysql> CREATE TABLE PriceAuditLog (
-> LogID INT PRIMARY KEY AUTO_INCREMENT,
-> BookID INT,
-> OldPrice DECIMAL(8,2),
-> NewPrice DECIMAL(8,2),
-> ChangedOn DATETIME DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.16 sec)

mysql> DESCRIBE PRICEAUDITLOG;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| LogID | int | NO | PRI | NULL | auto_increment |
| BookID | int | YES | | NULL | |
| OldPrice | decimal(8,2) | YES | | NULL | |
| NewPrice | decimal(8,2) | YES | | NULL | |
| ChangedOn | datetime | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.05 sec)

mysql> |
```

- **BEFORE INSERT TRIGGER (ENFORCE A BUSINESS RULE)**

Business rule: a book's price can never be inserted as zero or negative.

SYNTAX:

DELIMITER //

```
CREATE TRIGGER trigger_name
```

```
BEFORE INSERT ON table_name

FOR EACH ROW

BEGIN

    IF NEW.column condition THEN

        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'message';

    END IF;

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE TRIGGER trg_check_price

    BEFORE INSERT ON Book

    FOR EACH ROW

BEGIN

    IF NEW.Price <= 0 THEN

        SIGNAL SQLSTATE '45000'

        SET MESSAGE_TEXT = 'Price must be greater than zero.';

    END IF;

END //

DELIMITER ;

INSERT INTO Book VALUES (12, 'Free Sample Book', 'Unknown', 0,

'978-93-100-0013', NULL, 1, 1, 1);

INSERT INTO Book VALUES (12, 'City Skylines', 'Nikhil Rao', 430.00,

'978-93-100-0013', NULL, 5, 2, 1);
```

```

Command Prompt - mysql -u x
Windows PowerShell
Windows PowerShell

-> OldPrice DECIMAL(8,2),
-> NewPrice DECIMAL(8,2),
-> ChangedOn DATETIME DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.16 sec)

mysql> DESCRIBE PRICEAUDITLOG;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| LogID | int | NO | PRI | NULL | auto_increment |
| BookID | int | YES | | NULL | |
| OldPrice | decimal(8,2) | YES | | NULL | |
| NewPrice | decimal(8,2) | YES | | NULL | |
| ChangedOn | datetime | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.05 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER trg_check_price
-> BEFORE INSERT ON Book
-> FOR EACH ROW
-> BEGIN
-> IF NEW.Price <= 0 THEN
-> SIGNAL SQLSTATE '45000'
-> SET MESSAGE_TEXT = 'Price must be greater than zero.';
-> END IF;
-> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Book VALUES (12, 'Free Sample Book', 'Unknown', 0, '978-93-100-0013', NULL, 1, 1, 1);
ERROR 1644 (45000): Price must be greater than zero.
mysql> INSERT INTO Book VALUES (12, 'City Skylines', 'Nikhil Rao', 430.00, '978-93-100-0013', NULL, 5, 2, 1);
Query OK, 1 row affected (0.01 sec)

mysql> |

```

● AFTER UPDATE TRIGGER (AUTOMATIC AUDIT LOGGING)

SYNTAX:

DELIMITER //

CREATE TRIGGER trigger_name

AFTER UPDATE ON table_name

FOR EACH ROW

BEGIN

INSERT INTO audit_table VALUES (... , OLD.column, NEW.column, ...);

END //

DELIMITER ;

QUERY:

DELIMITER //

CREATE TRIGGER trg_log_price_change

AFTER UPDATE ON Book

FOR EACH ROW

BEGIN

IF OLD.Price <> NEW.Price THEN

INSERT INTO PriceAuditLog (BookID, OldPrice, NewPrice)

VALUES (OLD.BookID, OLD.Price, NEW.Price);

END IF;

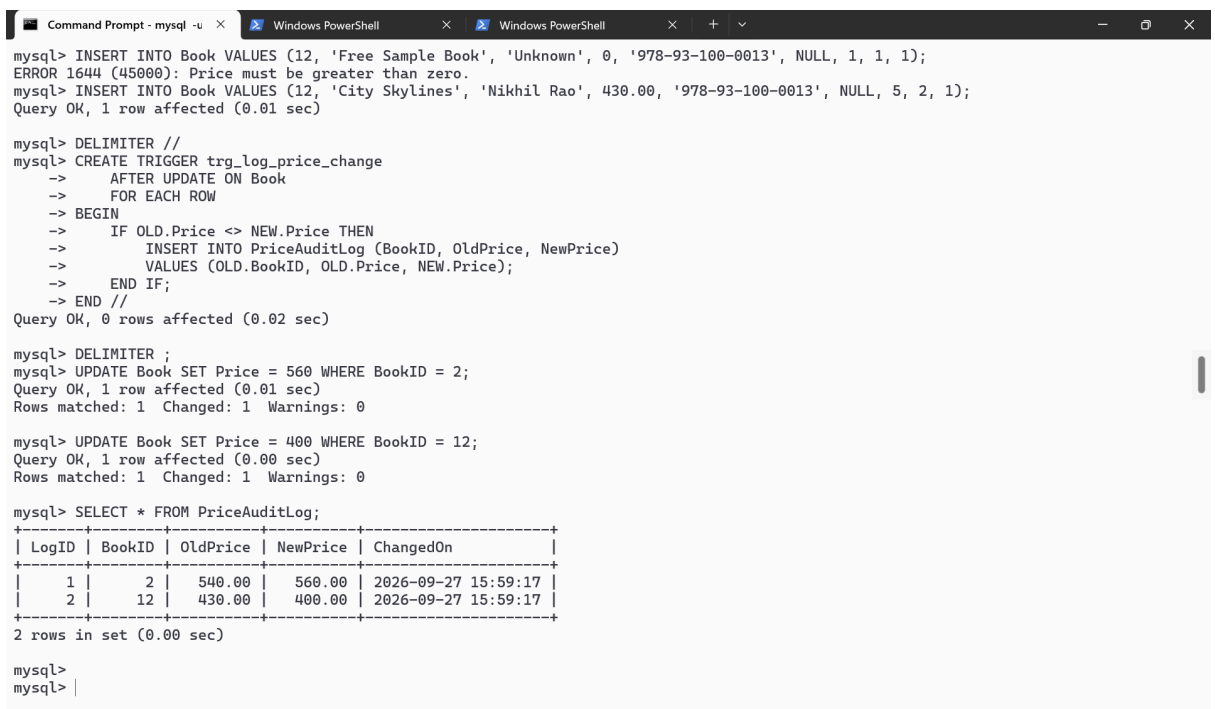
END //

DELIMITER ;

UPDATE Book SET Price = 560 WHERE BookID = 2;

UPDATE Book SET Price = 400 WHERE BookID = 12;

SELECT * FROM PriceAuditLog;



```

mysql> INSERT INTO Book VALUES (12, 'Free Sample Book', 'Unknown', 0, '978-93-100-0013', NULL, 1, 1, 1);
ERROR 1644 (45000): Price must be greater than zero.
mysql> INSERT INTO Book VALUES (12, 'City Skylines', 'Nikhil Rao', 430.00, '978-93-100-0013', NULL, 5, 2, 1);
Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER trg_log_price_change
-> AFTER UPDATE ON Book
-> FOR EACH ROW
-> BEGIN
-> IF OLD.Price <> NEW.Price THEN
-> INSERT INTO PriceAuditLog (BookID, OldPrice, NewPrice)
-> VALUES (OLD.BookID, OLD.Price, NEW.Price);
-> END IF;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> UPDATE Book SET Price = 560 WHERE BookID = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Book SET Price = 400 WHERE BookID = 12;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM PriceAuditLog;
+-----+-----+-----+-----+-----+
| LogID | BookID | OldPrice | NewPrice | ChangedOn |
+-----+-----+-----+-----+-----+
| 1 | 2 | 540.00 | 560.00 | 2026-09-27 15:59:17 |
| 2 | 12 | 430.00 | 400.00 | 2026-09-27 15:59:17 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql>

```

11. EXPLICIT AND IMPLICIT CURSORS

- IMPLICIT CURSOR EQUIVALENT: ROW_COUNT()

SYNTAX:

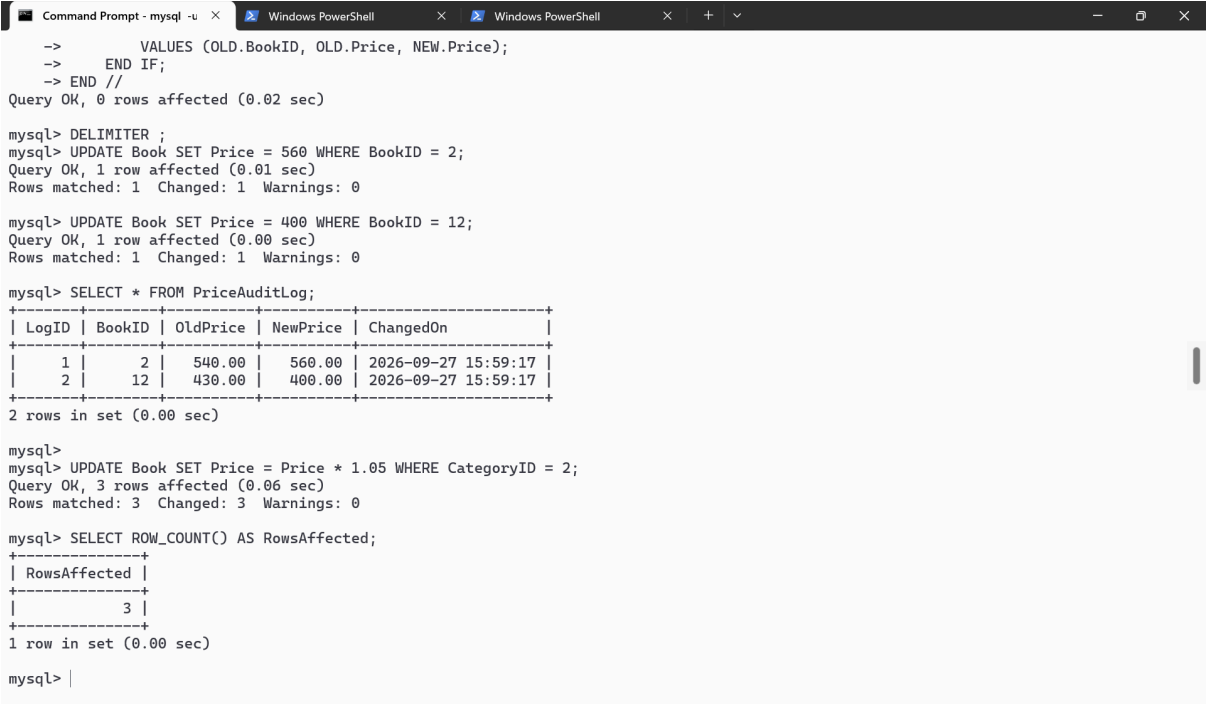
UPDATE table_name SET column = expression WHERE condition;

```
SELECT ROW_COUNT() AS RowsAffected;
```

QUERY:

```
UPDATE Book SET Price = Price * 1.05 WHERE CategoryID = 2;
```

```
SELECT ROW_COUNT() AS RowsAffected;
```



```

-> VALUES (OLD.BookID, OLD.Price, NEW.Price);
-> END IF;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> UPDATE Book SET Price = 560 WHERE BookID = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE Book SET Price = 400 WHERE BookID = 12;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM PriceAuditLog;
+-----+-----+-----+-----+-----+
| LogID | BookID | OldPrice | NewPrice | ChangedOn |
+-----+-----+-----+-----+-----+
| 1 | 2 | 540.00 | 560.00 | 2026-09-27 15:59:17 |
| 2 | 12 | 430.00 | 400.00 | 2026-09-27 15:59:17 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
mysql> UPDATE Book SET Price = Price * 1.05 WHERE CategoryID = 2;
Query OK, 3 rows affected (0.06 sec)
Rows matched: 3 Changed: 3 Warnings: 0

mysql> SELECT ROW_COUNT() AS RowsAffected;
+-----+
| RowsAffected |
+-----+
| 3 |
+-----+
1 row in set (0.00 sec)

mysql>

```

● EXPLICIT CURSOR (PROCESSING MULTIPLE ROWS)

SYNTAX:

```
DELIMITER //
```

```
CREATE PROCEDURE procedure_name()
```

```
BEGIN
```

```
    DECLARE done INT DEFAULT FALSE;
```

```
    DECLARE v_col1 datatype;
```

```
    DECLARE v_col2 datatype;
```

```
    DECLARE cur CURSOR FOR SELECT col1, col2 FROM table_name WHERE
condition;
```

```
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
```

```
OPEN cur;

read_loop: LOOP

    FETCH cur INTO v_col1, v_col2;

    IF done THEN

        LEAVE read_loop;

    END IF;

    -- process v_col1, v_col2

END LOOP;

CLOSE cur;

END //
```

QUERY:

```
DELIMITER //
```

```
CREATE PROCEDURE prog_FictionLabels()

BEGIN

    DECLARE done INT DEFAULT FALSE;

    DECLARE v_title VARCHAR(100);

    DECLARE v_price DECIMAL(8,2);

    DECLARE cur CURSOR FOR SELECT Title, Price FROM Book WHERE

CategoryID = 1;

    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

    DROP TEMPORARY TABLE IF EXISTS FictionLabels;

    CREATE TEMPORARY TABLE FictionLabels (Title VARCHAR(100), Price

DECIMAL(8,2), Label VARCHAR(20));
```

```
OPEN cur;

read_loop: LOOP

    FETCH cur INTO v_title, v_price;

    IF done THEN

        LEAVE read_loop;

    END IF;

    IF v_price > 400 THEN

        INSERT INTO FictionLabels VALUES (v_title, v_price, 'Bestseller');

    ELSE

        INSERT INTO FictionLabels VALUES (v_title, v_price, 'Regular');

    END IF;

END LOOP;

CLOSE cur;

SELECT * FROM FictionLabels;

END //

DELIMITER ;

CALL prog_FictionLabels();
```

```

Command Prompt - mysql -u x  Windows PowerShell  Windows PowerShell  x + v
--> CREATE TEMPORARY TABLE FictionLabels (Title VARCHAR(100), Price DECIMAL(8,2), Label VARCHAR(20));
-->
--> OPEN cur;
--> read_loop: LOOP
-->     FETCH cur INTO v_title, v_price;
-->     IF done THEN
-->         LEAVE read_loop;
-->     END IF;
-->
-->     IF v_price > 400 THEN
-->         INSERT INTO FictionLabels VALUES (v_title, v_price, 'Bestseller');
-->     ELSE
-->         INSERT INTO FictionLabels VALUES (v_title, v_price, 'Regular');
-->     END IF;
--> END LOOP;
--> CLOSE cur;
-->
--> SELECT * FROM FictionLabels;
--> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL prog_FictionLabels();
+-----+-----+-----+
| Title           | Price | Label   |
+-----+-----+-----+
| The Silent Peak | 493.50 | Bestseller |
| Midnight in Kerala | 651.00 | Bestseller |
| Whispers of Rajasthan | 357.00 | Regular |
| Poems of the Coast | 378.00 | Regular |
+-----+-----+-----+
4 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> |

```

● PARAMETERIZED CURSOR (REUSABLE ACROSS ANY CATEGORY)

SYNTAX:

DELIMITER //

CREATE PROCEDURE procedure_name(IN p_param datatype)

BEGIN

 DECLARE done INT DEFAULT FALSE;

 DECLARE v_col1 datatype;

 DECLARE v_total INT DEFAULT 0;

 DECLARE cur CURSOR FOR SELECT col1 FROM table_name WHERE column
= p_param;

 DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

 OPEN cur;

 read_loop: LOOP

```
    FETCH cur INTO v_col1;

    IF done THEN

        LEAVE read_loop;

    END IF;

    SET v_total = v_total + v_col1;

END LOOP;

CLOSE cur;

SELECT v_total AS Total;

END //
```

QUERY:

```
DELIMITER //
```

```
CREATE PROCEDURE prog_TotalStockByCategory(IN p_catid INT)

BEGIN

    DECLARE done INT DEFAULT FALSE;

    DECLARE v_stock INT;

    DECLARE v_total INT DEFAULT 0;

    DECLARE cur CURSOR FOR SELECT Stock FROM Book WHERE CategoryID

= p_catid;

    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;

    OPEN cur;

    read_loop: LOOP

        FETCH cur INTO v_stock;

        IF done THEN
```

```

        LEAVE read_loop;

    END IF;

    SET v_total = v_total + v_stock;

END LOOP;

CLOSE cur;

SELECT p_catid AS CategoryID, v_total AS TotalStock;

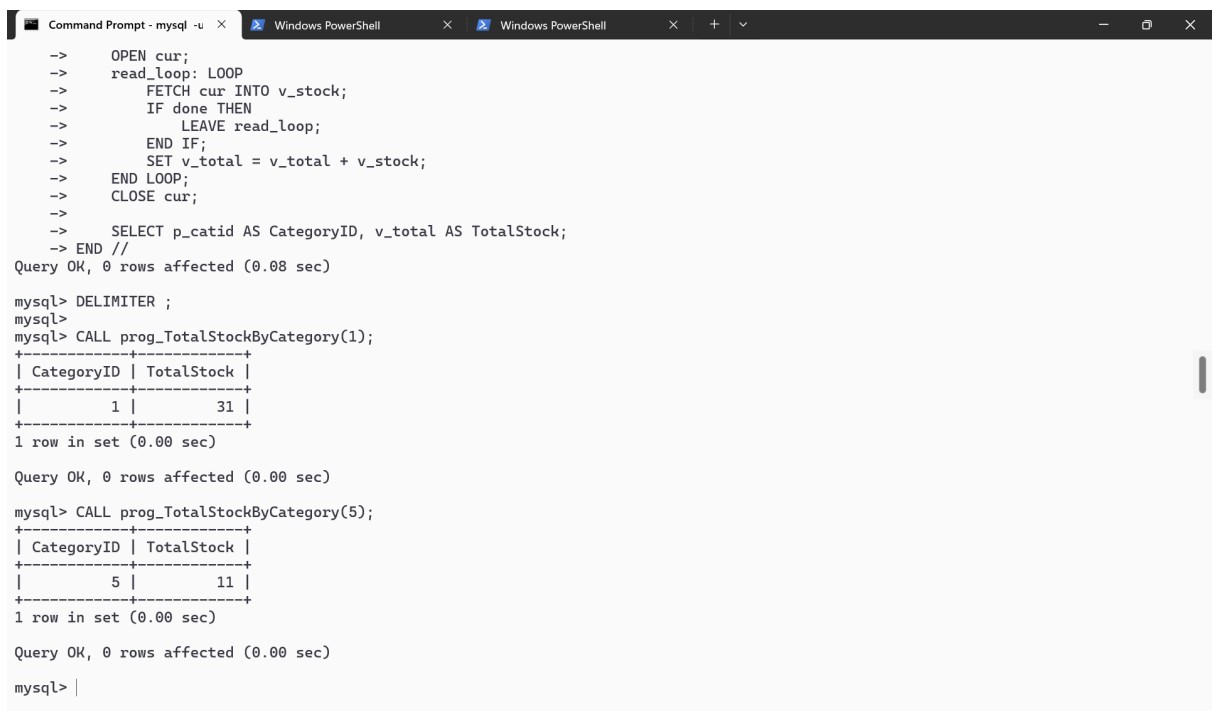
END //

DELIMITER ;

CALL prog_TotalStockByCategory(1);

CALL prog_TotalStockByCategory(5);

```



```

Command Prompt - mysql -u x  Windows PowerShell  Windows PowerShell
-> OPEN cur;
-> read_loop: LOOP
->     FETCH cur INTO v_stock;
->     IF done THEN
->         LEAVE read_loop;
->     END IF;
->     SET v_total = v_total + v_stock;
-> END LOOP;
-> CLOSE cur;
->
-> SELECT p_catid AS CategoryID, v_total AS TotalStock;
-> END //
Query OK, 0 rows affected (0.08 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL prog_TotalStockByCategory(1);
+-----+-----+
| CategoryID | TotalStock |
+-----+-----+
|          1 |          31 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> CALL prog_TotalStockByCategory(5);
+-----+-----+
| CategoryID | TotalStock |
+-----+-----+
|          5 |          11 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> |

```

CONCLUSION

The Library Management System successfully integrates database design, SQL, and MySQL's procedural features into a single practical application. The system manages books, categories, publishers, members, and issue records while maintaining data integrity throughout. The project demonstrates core database concepts including constraints, joins, aggregate functions, subqueries, views, transaction control, stored procedures, functions, triggers, and cursors, all implemented and verified on a live, continuously evolving database.

